Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 35)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 1 . 1 Explain the main advantage of GM over conventional selective breeding.

[2 marks]

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0 1 . 2 For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops .....

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Roundup Ready soya .....

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Golden rice .....

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0 1 . 3 Explain **four** concerns about GM crops.

[4 marks]

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0 1 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

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0 2 Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	**1900**	**1930**	**1960**	**1986**	**2020**
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

0 2 . 1 Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer ..... %

0 2 . 2 Explain why whale populations were so easily over-exploited.

[3 marks]

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0 2 . 3 Describe how the International Whaling Commission regulates whaling.

[4 marks]

0 2 . 4 Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

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0 3 Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 3 . 1 Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

0 3 . 2 Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

0 3 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

0 3 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

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0 4 . 1 Explain why tropical rainforests have such high biodiversity.

[4 marks]

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0 4 . 2 Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

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0 4 . 3 Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

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0 4 . 4 Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

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0 5 . 1 Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

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0 5 . 2 Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

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0 5 . 3 Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

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0 5 . 4 Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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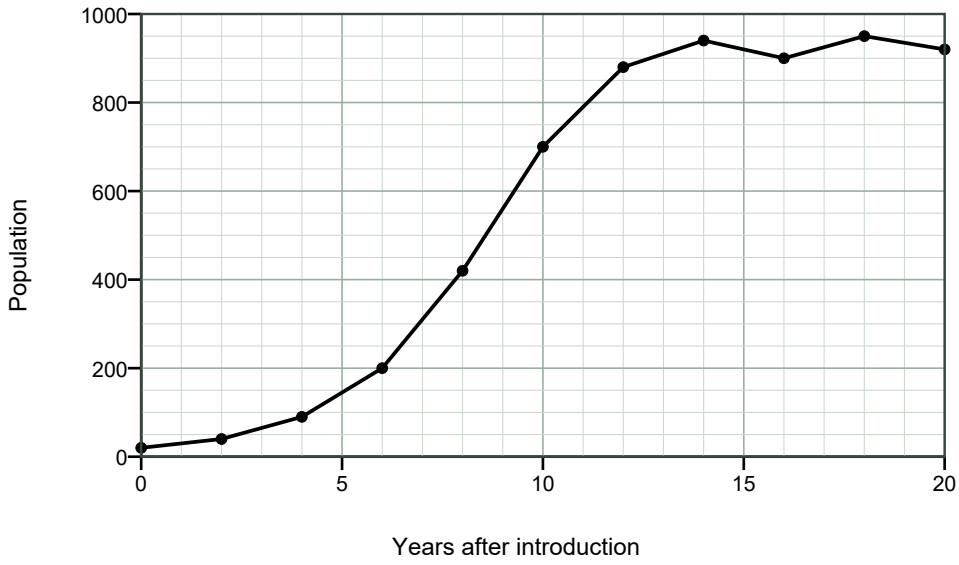
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0 6 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 6 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

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0 6 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

0 6 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

0 6 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 7 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed

Sea lice and disease

Escapes affecting wild gene pools

Organic waste pollution

0 7 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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0 8 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 8 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 8 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 9 . 1 For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

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Food .....

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Fashion .....

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Entertainment / pets .....

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Furniture and ornaments .....

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Traditional medicines .....

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0 9 . 2 Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

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0 9 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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1 0 . 1 Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **rewilding**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS